high-power radio stations (300kW or 200kW) can have an antenna of about 20 metres (see Fig. 1). For low-power areas, one should have a longer antenna that can go up to 36 metres (120 feet). If space is a problem, the antenna wire can be arranged in L or U shape around the house at the maximum possible height (say, 6 to 9 metres), ensuring that the wire does not touch a wall.

Components required for a crystal radio are:

1. A good aerial/antenna made of 18 to 36 metres (60 to 120 feet) long enameled copper wire (18 or 20 SWG), or insulated copper wire (hook-up wire: 14/36) at a height of 6 to 9 metres (20 to 30 feet).

2. A good earth connected to a metallic water pipe through a tap. In recent days, PVC pipes are used for water lines instead of metal pipes, so double-check the earth connection if your crystal radio does not work. In that case the earth point of an electrical wall socket can be used with due care. (***WARNING.*** Please consult an experienced electrical engineer before using earth connection from a wall socket as improper connection can be dangerous and even fatal!)

Fig. 2: Modified plug for earth connection

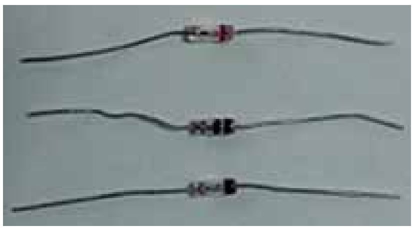

Fig. 3: Germanium diodes

To use an electrical wall socket for the earth, take a 3-pin plug and open it by unscrewing the retaining screws. Remove the two small pins meant for 'Line' and 'Neutral' and discard them. Connect one or two metres of insulated copper wire to the bigger and thick pin meant for 'Earth' connection. Now close the plug using the screws. The other end of the wire is to be connected to the earth point of the crystal radio, and the modified plug needs to be inserted into the wall socket (see the photos of the modified plug in Fig. 2).

3. A 1N34A or equivalent germanium diode or Schottky diode 1N5817/1N5818 (which would require much longer antenna). Most diodes sold as germanium diode 1N34A (refer Fig. 3) in electronics markets do not have marking on them, and many of them are not actually 1N34A and so do not work in a crystal radio. The diode having a forward voltage (Vf) of less than 300mV works with crystal radios and any diode having its Vf more than 300mV will not work. Vf of 225mV to 250mV is the best choice. You can check the Vf of a diode using a digital multimeter that has Diode Check facility.

Set the selector switch of the multimeter to the Diode Test symbol. Connect black probe of the multimeter to cathode terminal of the diode. The cathode terminal is on the side with a black or red or any other coloured band. Connect the red probe to the other terminal (anode). The reading on the meter will show the forward voltage. If the forward voltage is less than 300mV, you can use this diode for your crystal radio.

4. A good high-impedance headphone having 2,000 to 4,000 ohms DC resistance, or landline telephone earphone having 150 to 300 ohms DC resistance (refer Fig. 4), or an earphone used for MP3 players or cellphones having 16 to 32 ohms resistance with an impedance-matching transformer (refer Figs 5 and 6). Nowadays, it is difficult to obtain high-impedance headphones. If you are lucky, you may get it in junk

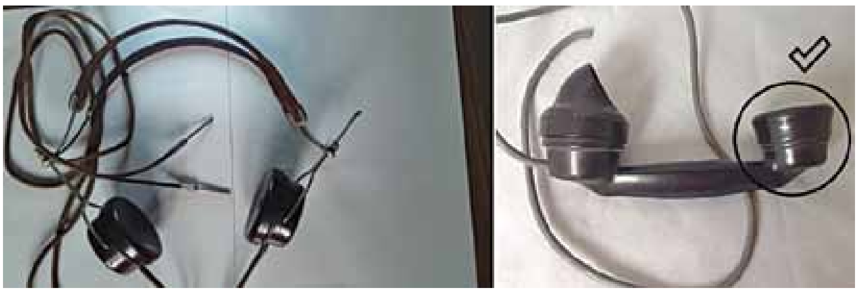

Fig. 4: High-impedance headphone (2,000 to 4,000 ohms)

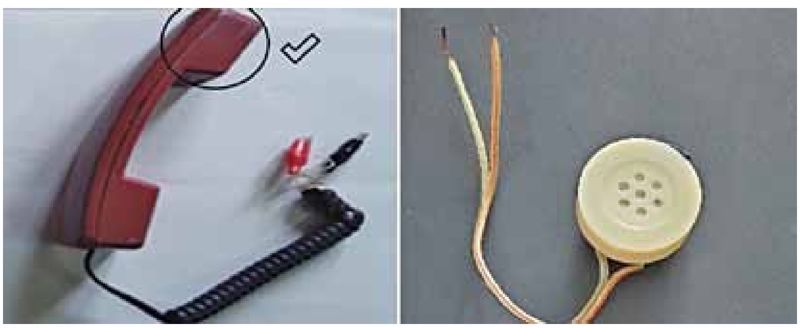

Fig. 5: Pushbutton telephone handsets used in crystal radio (150 ohms)

Fig. 6: Impedance-matching transformer and earphones